IPv4 vs IPv6

FEATURE	Internet Protocol version 4 (IPv4)	Internet Protocol version 6 (IPv6)
Initial Development	1981	1999
Address Size	32-bit number (4 bytes)	128-bit number (16 bytes)
Address Format	Dotted Decimal Notation	Hexadecimal Notation with Colons
Prefix Notation	192.168.0.0/24	2001:db8::/32
Number of Addresses	2^{32} (4.3 billion)	2^{32} (340 undecillion)
NAT Requirement	Commonly required due to address scarcity	Typically not need (end to end connectivity)
Security	IPsec optional	IPsec built in
Auto-configuration	DHCP required	Stateless (SLAAC) built-in

IPv4 Address Structure

Example: 192.168.254.1 (dotted decimal)

192	168	254	1

- Each number (octet) ranges from 0-255
- Each octed = 8 bits (1 byte)
- Total length: 32 bits (4 bytes)
- Binary representation

IPv6 Address Structure

Example: 2001:0db8:85a3::8a2e:0370:7334

2001	0db8	85a3	::	8a2e	0370	7334	

- Eight 16-bit groups (hextets) seperated by colons
- Each hextet ranges from 0000-FFFF (in hexadecimal)
- Zero compression "::" replaces consecutive zero groups
- Leading zeros in a group can be ommited (0db8 = db8)

Protocol Header Comparison

Version	IHL	Type of Service	Total Length
Identification	Flags	Fragment Offset	
Time to Live (TTL)			
Protocol			
Header Checksum	Options & Padding		
Source IP Address			
Destination IP Address			

Version	Traffic Class	Flow Label	(FLow Control)
Payload Length			
Next Header	Hop Limit		
Source IPv6 Address (16 bytes)			
Destination IPv6 Address (16 bytes)			